

Atomistic understanding of the network dilation anomaly in ion-exchanged glass

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ABSTRACT

Chemically strengthened glasses are of increasing technological importance for personal electronic devices, tablet computers, and a variety of other applications. However, there are many unexplained phenomena associated with the physics of the ion exchange process used for strengthening. One of the most puzzling of these is the anomalous behavior of the network dilation coefficient, i.e., the parameter governing the amount of linear strain of the glass per unit of alkali ions exchanged, which is inevitably a factor of 2–4 higher for as-melted glasses as compared to chemically strengthened versions of the same glass compositions prepared via ion exchange. In this paper, we investigate the atomistic origin of this discrepancy between as-melted and ion-exchanged glasses based on molecular dynamics simulations of a series of alkali tetrasilicate glasses, viz., $x\text{Na}_2\text{O} \cdot (20-x)\text{K}_2\text{O} \cdot 80\text{SiO}_2$ (mol%). The network dilation coefficient of the ion-exchanged glasses is dependent on composition and ranges from 30% to 54% of the ideal value obtained from the as-melted glasses. This anomalous behavior of the network dilation coefficient is explained in terms of different local environments between sodium and potassium sites in the glass network and a two-stage relaxation process of the local potassium environment after ion exchange.

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1. Introduction

From glass windows and displays to lenses and optical fiber, glassy materials have proved to be one of the key enablers of modern civilization [1–3]. Many new applications for glass are expected to address current global issues regarding energy [4,5], medicine [6–8], and the environment [9,10]. However, one of the main limitations of glassy materials is their brittle fracture behavior [11,12]. This is especially problematic in the presence of surface flaws, which act to concentrate applied tensile stress onto localized regions of the material [13]. Such surface flaws can act to lower the practical strength of a glass product to just a small fraction of its ideal theoretical strength.

Significant enhancement of the practical strength of glass products can be achieved through the ion exchange process [14–17]. Ion exchange strengthening of glass was originally developed by Kistler [18] in 1962 but has only recently become a widespread technique applied for commercial glass products. The chemical strengthening of glass through ion exchange has several advantages compared to conventional thermal tempering. Significantly higher compressive stresses can be achieved through chemical strengthening. Also, the ion exchange process enables strengthening of glass products with complex geometries [14,15].

During the ion exchange process, a glass containing some concentration of an alkali oxide (typically Li_2O or Na_2O) is immersed in a molten salt bath consisting of a larger alkali. Typically the salt bath is chosen as NaNO_3 for a Li-containing glass or KNO_3 for a Na-containing glass. During the salt bath immersion, smaller cations from the glass diffuse into the salt bath and are replaced by larger cations from the salt. By “stuffing” these larger cations into sites previously occupied by a smaller species, a layer of compressive stress develops near the surface of the glass. This compressive stress acts to inhibit crack formation and propagation, significantly increasing the damage resistance of the glass [14,15].

Neglecting stress relaxation effects, the stress profile obtained via the ion exchange process is given by

$$\sigma(z) = -\frac{BE}{1-\nu} [C(z) - C_{\text{avg}}], \quad (1)$$

where E is Young's modulus, ν is Poisson's ratio, and $C(z)$ is the local concentration of the substituting cation at a penetration depth z in the glass. In order for Eq. (1) to satisfy the force balance condition, the average concentration of the substituting cations, C_{avg} , must be subtracted from $C(z)$ such that the integral of $\sigma(z)$ over the complete thickness of the glass is zero. Hence, the negative (compressive) stress near the surface of the glass is exactly compensated by a positive (tensile) stress in the interior [19].

The parameter B in Eq. (1) is the linear network dilation coefficient (LNDC) [19–22], also known as the “Cooper coefficient” in honor of the

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late Prof. Alfred R. Cooper, one of the pioneers in understanding the physics of the ion exchange process. The concept of the network dilation coefficient was originally proposed by Richmond et al. [23] for the calculation of stresses due to nitrogenation of metals. It can be considered as a quantity analogous to the thermal expansion coefficient. Whereas the linear thermal expansion coefficient defines the linear strain of a material per unit change in temperature, the linear network dilation coefficient B gives the linear strain per unit change in alkali concentration, viz., [19]

$$B = \frac{1}{3} \left(\frac{1}{V} \frac{\partial V}{\partial C} \right) = \frac{1}{3} \frac{\partial \ln V}{\partial C}, \quad (2)$$

where V is the molar volume of the glass.

One of the mysteries of the ion exchange process relates to the anomalous behavior of the LNDC. Specifically, when Eq. (2) is evaluated by measuring the difference in molar volume of as-melted (unexchanged) glasses across sodium and potassium end members, the value of B is always a factor of 2–4 higher than that deduced from ion-exchanged glasses using Eq. (1) [15], [22]. In the latter case, the value of B is calculated using measured values of E , ν , $\alpha(0)$, $C(0)$, and C_{avg} . This factor of 2–4 persists even in the limit of short-time ion exchange experiments and therefore cannot be explained as a result of stress relaxation. From a technological point of view, the reduced value of B for ion-exchanged glasses is unfortunate since it leads to a compressive stress that is significantly lower than what would be expected based solely on as-melted versions of the same glass.

The objective of this paper is to develop an atomistic understanding of this anomalous behavior of the LNDC in as-melted versus ion-exchanged glasses. We address this issue through molecular dynamics simulations of a series of alkali tetrasilicate glasses spanning from sodium to potassium end members. From our molecular dynamics simulations, the LNDC of the “as-melted” glasses is 1.8 to 3.3 times larger than that of ion-exchanged glasses, where K atoms are substituted for Na. We explain the origin of this discrepancy in terms of differences in the local structure between sodium and potassium sites and a two-stage relaxation process of the local potassium environment in the ion-exchanged glasses.

2. Modeling procedure

We model eleven different alkali silicate compositions with a fixed SiO_2 concentration of 80 mol% and varying the $[\text{Na}_2\text{O}]/[\text{K}_2\text{O}]$ ratio between pure Na_2O and pure K_2O end members, i.e., $x\text{Na}_2\text{O} \cdot (20-x)\text{K}_2\text{O} \cdot 80\text{SiO}_2$ (mol%), where $x = 0, 2, 4, \dots, 20$. The initial structure for each composition is obtained by placing ~20,000 atoms at random positions in a cubic simulation cell. The large number of atoms is chosen so as to minimize errors in our calculated results. The initial dimensions of the cell are calculated from a neural network-based density model for silicate glasses, which is fit to experimental data in literature [24]. We apply periodic boundary conditions along all three planes. We model the “as-melted” glass structures through stepwise quenching starting at high temperature, with combined Metropolis Monte Carlo (MC) [25] and molecular dynamics (MD) runs at each temperature. The temperature begins at 4200 K and decreases in steps of 300 K down to 900 K. A final decrease of 200 K brings the glass to its final temperature of 700 K, which corresponds to a typical laboratory temperature for ion exchange. At each new temperature in the simulation, we calculate 100 million trial moves using MC under isobaric conditions (i.e., NPT ensemble at atmospheric pressure). The MC runs are conducted using an in-house code, Molsym [26,27], using a target acceptance ratio of 0.35 for both atomic displacements and cell volume changes. Equilibration during the MC runs is monitored based on relaxation of the total potential energy and the simulation cell volume.

The output structures from the MC runs are used as input for the MD simulations, which are conducted using GROMACS [28]. Each MD run is performed under constant volume (i.e., NVT) conditions for 600 ps using a time step of 1 fs. The output from the MD simulations is used as input for the subsequent round of simulations at the next temperature. At the final temperature of 700 K, we run another MD simulation in the NPT ensemble for 8 ns to calculate the final glass structure and molar volume of the system. We have found that a combination of MC and MD runs such as this provides the most efficient and accurate procedure for preparing glass structures through atomistic level simulation [29,30]. All of the simulations are performed using the Pedone set of interatomic potentials [31]. Electrostatic interactions are calculated using the Wolf method [32] in the MC simulations and using particle mesh Ewald (PME) [33] in the MD engine.

After the “as-melted” glasses are equilibrated at 700 K, the “ion-exchanged” glasses are prepared by a direct substitution of Na with K atoms. In each glass we consider a complete exchange of all Na atoms with K. This substitution is followed by an MD simulation for 6 ns in the NVT ensemble followed by another 8 ns of equilibration in the NPT ensemble. The temperature is held constant at 700 K since this corresponds to a typical salt bath temperature used during the laboratory ion exchange process. The calculation of the LNDC is based on the system’s final molar volume before and after the Na exchange with K. Calculated values of molar volume are accurate within roughly $\pm 1\%$.

3. Results

The final molar volumes of the as-melted and ion-exchanged glasses are plotted in Fig. 1 as a function of the sodium concentration in the as-melted glass. The as-melted glasses display a nearly perfect linear scaling of molar volume with concentration, leading to a constant value of the network dilation coefficient, as indicated by the dashed line in Fig. 2.

After the as-melted glasses undergo ion exchange, their composition is identical to that of the potassium end member of the as-melted glass series. Hence, if the ion-exchanged glasses exhibited the network dilation expected from the as-melted glasses, then the molar volume of all the ion-exchanged glasses would be the same, i.e., identical to that of the as-melted potassium end member. However, Fig. 1 shows that the molar volume of the ion-exchanged glasses is (a) dependent on the initial glass composition and (b) always less than that of the as-melted potassium glass. This result is indicative

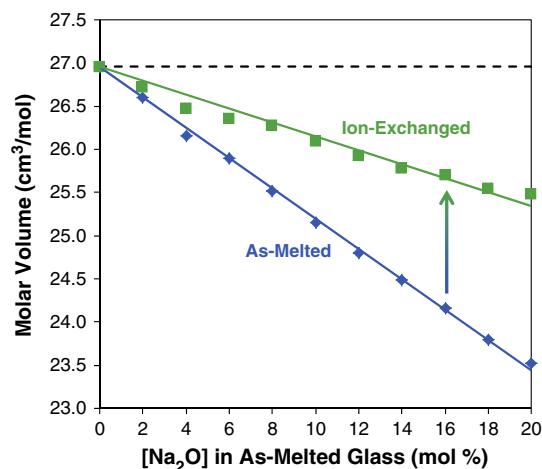


Fig. 1. Composition dependence of molar volume for the as-melted and ion-exchanged glasses in the $x\text{Na}_2\text{O} \cdot (20-x)\text{K}_2\text{O} \cdot 80\text{SiO}_2$ (mol%) system at 700 K.

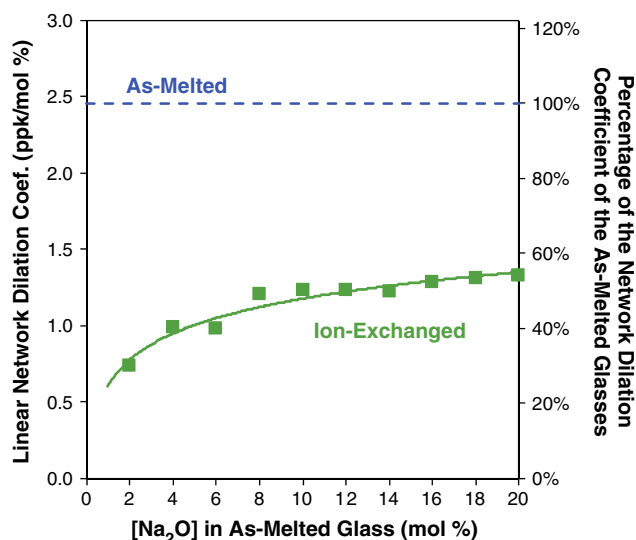


Fig. 2. Linear network dilation coefficient (LNDC) of the as-melted versus ion-exchanged glasses. The LNDC of the ion-exchanged glasses is composition-dependent and ranges from 30% to 54% of the ideal value of the as-melted glasses.

of the network dilation anomaly, i.e., a lower LNDC obtained via ion exchange compared to what is found from as-melted glasses.

The calculated network dilation coefficients of the ion-exchanged glasses are plotted in Fig. 2. Here it is shown that the LNDC found during ion exchange ranges from 30% to 54% of the ideal coefficient calculated from the as-melted glasses. This result is consistent with the experimental observation that the LNDC is roughly a factor of 2–4 lower than that of corresponding as-melted glasses [15], [22] and also consistent with recent molecular dynamics simulations presented by Kreski et al. [34] Fig. 2 also shows a clear composition dependence of the network dilation coefficient for the ion-exchanged glasses, with a lower coefficient obtained for glasses starting with a lower concentration of Na_2O . This result indicates a greater difficulty for potassium to strain the glass when there are fewer initial sodium sites in the system.

In order to understand the atomistic origin of this network dilation anomaly we perform a detailed analysis of the simulated glass structures. We find that the main structural difference between the as-melted potassium tetrasilicate glass and the analogous ion-exchanged glasses is in the local environment around potassium. Fig. 3 shows the coordination number distribution of oxygen around alkali atoms for the $[\text{Na}_2\text{O}] = 10$ and 18 mol% glasses before and after ion exchange. The K–O coordination number of ~9 in the as-melted glasses is in good agreement with XAFS measurements for potassium tetrasilicate [35], and the Na–O coordination number of ~6 is in agreement with that found in *ab initio* simulations of sodium tetrasilicate [36]. Here the coordination numbers are calculated for the first coordination sphere of oxygen, i.e., including the first full peak of the K–O radial distribution function. In the ion-exchanged glasses, where the potassium atoms are substituted into sodium sites, there is a significantly lower K–O coordination number compared to that in the original as-melted environment.

The average K–O coordination numbers of the as-melted and ion-exchanged glasses are plotted in Fig. 4(a) for all of the compositions under investigation. This figure shows a clear downward shift of the average K–O coordination number in the ion-exchanged glasses compared to the K–O distribution in the original as-melted glasses. However, the average K–O distribution in the ion-exchanged glasses is still higher than what is calculated for the as-melted potassium end member composition, depicted also in Fig. 4(b). Comparing Fig. 3 with Fig. 4(b), one can see that the K–O coordination distributions of the ion-exchanged glasses take on intermediate values between those of the original as-melted glasses (with a higher average coordination

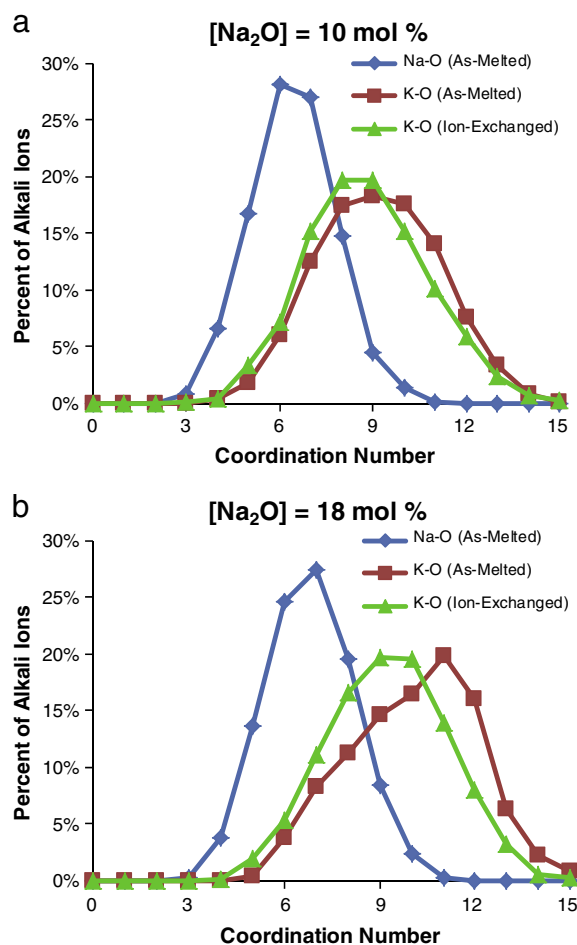


Fig. 3. Distribution of oxygen coordination number around sodium and potassium in the (a) $[\text{Na}_2\text{O}] = 10$ mol % and (b) $[\text{Na}_2\text{O}] = 18$ mol % glasses. The K–O distribution is shifted to lower coordination in the ion-exchanged glasses, an effect that is especially pronounced at high levels of $[\text{Na}_2\text{O}]$ in the initial glass.

number) and that of the ideal as-melted potassium tetrasilicate end member (with a lower average coordination number).

While there are significant changes in the local environment around potassium between the as-melted and ion-exchanged glasses, Fig. 5 shows that there is no difference in the Q^n speciation of silica in these glasses, where n indicates the number of bridging oxygens per silica tetrahedron. This figure indicates that the distribution of non-bridging oxygens is independent of the particular fraction of sodium and potassium in the alkali tetrasilicate glasses. The Q^n distribution is also found to be independent of whether or not the glasses are ion-exchanged. Thus, the local environment around silicon apparently plays no role in the network dilation anomaly.

4. Discussion

Our molecular dynamics simulations reveal that the network dilation anomaly is a result of differences in the local alkali environments between as-melted and ion-exchanged glasses. In general, potassium ions prefer a higher oxygen coordination sphere compared to that of sodium, a result of the higher ionic radius of potassium. When potassium is substituted for sodium during the ion exchange process, it is forced into a smaller site with a lower coordination of oxygen. Local structural rearrangements quickly allow for a greater K–O coordination in the ion-exchanged glass, but without a fully commensurate increase in molar volume. As indicated in Fig. 4(a), the presence of sodium in the as-melted glasses actually encourages a higher coordination number of potassium compared to that of the sodium-free end

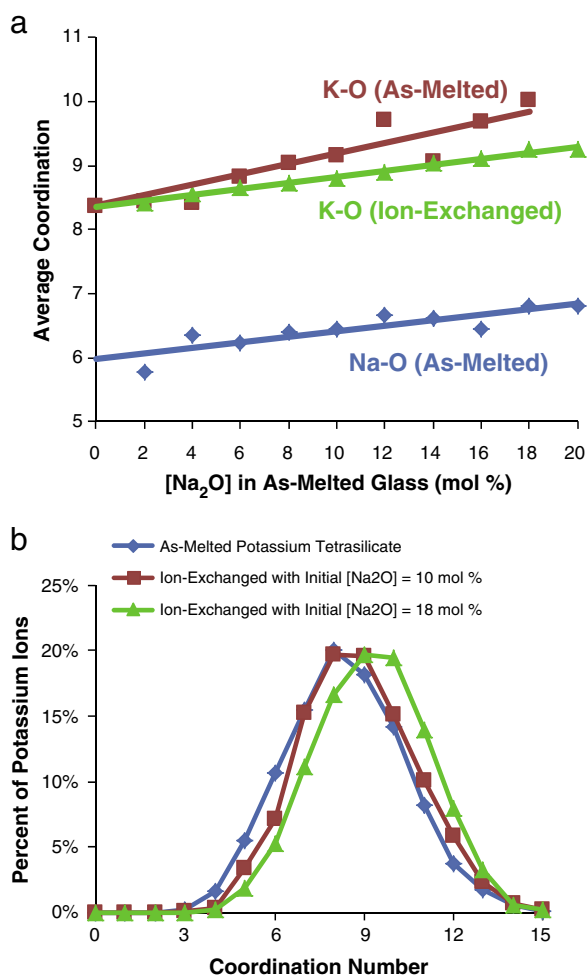


Fig. 4. (a) Average coordination of oxygen around Na and K in the as-melted and ion-exchanged glasses. (b) Distribution of oxygen coordination number around potassium in the as-melted potassium tetrasilicate glass, as well as the ion-exchanged glasses with initial [Na₂O] = 10 mol% and 18 mol%.

member. Thus, even before ion exchange, the preferred coordination number of potassium is higher in mixed alkali glasses than in the as-melted potassium tetrasilicate glass.

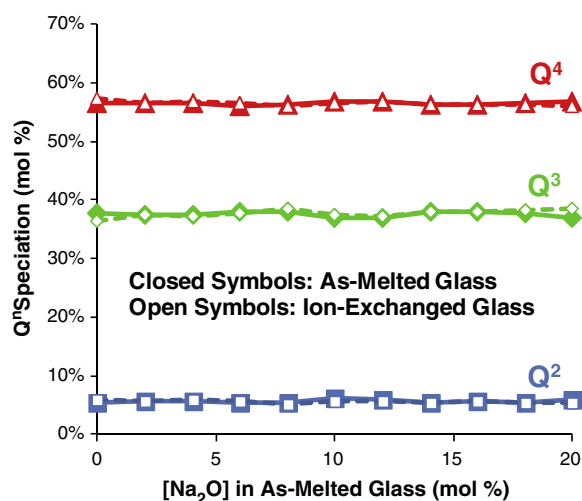


Fig. 5. Qⁿ speciation of silica in the as-melted (closed symbols) and ion-exchanged (open symbols) glasses. There is no discernable difference in the Qⁿ distribution due to the ion-exchange process.

When the potassium ions are substituted for sodium during the ion exchange process, the average K–O coordination number takes on an intermediate value between what would be expected for potassium in the initial sodium-containing glass and what is ultimately desired in the pure potassium end member. The relaxation of the local structure is apparently a two-stage process. An initial fast relaxation involves rearrangement of the local potassium environment toward what would be expected in the presence of sodium, i.e., in the as-melted mixed alkali glasses. A second, slower stage of relaxation involves gradual rearrangement of the structure toward the as-melted potassium end member, reflecting the change in composition of the glass due to the ion exchange process.

The network dilation anomaly is a direct result of this two-stage relaxation process. The strain that is actually achieved in an ion-exchanged glass is the result of the fast local rearrangement of atoms that occurs on the time scale of the salt bath treatment. The resulting strain from these initial structural rearrangements is what gives rise to the observed LNDC of the ion-exchanged glass. This value is significantly smaller compared to what would be obtained if the ideal structure of the as-melted potassium end member is achieved, i.e., given sufficiently long time to achieve further structural relaxation. The two-stage nature of this relaxation process therefore presents a limitation on the amount of strain that can be practically obtained during the ion exchange process. Since this strain is converted into stress via a constant of proportionality, $-E/(1-\nu)$, as in Eq. (1), this limitation on strain also limits the achievable compressive stress during ion exchange strengthening.

A consequence of this two-stage relaxation process is that it may be possible to overcome the network dilation anomaly by ion-exchanging the glass at higher temperatures, since this would provide additional thermal energy for relaxation of the local environment around potassium. However, the magnitude of stress relaxation would also increase at higher salt bath temperatures. While greater initial compressive stress may be installed due to the higher dilatational strain, the stress would be rapidly lost due to stress relaxation. Thus, such high temperature processing effectively defeats the purpose of the ion exchange process, since we wish to install elastic strain rather than plastic strain into the system.

Several previous authors have also invoked a low-temperature relaxation mechanism as a possible explanation of the network dilation anomaly [20], [37–41]. In these earlier works the effect is usually attributed to network plasticity, i.e., an irrecoverable deformation of the glass network on a laboratory time scale. Since stress buildup is the result of elastic strain only, the presence of plastic strain would lead to lower values of the network dilation coefficient as calculated from Eq. (1). Our current results show that the network dilation anomaly is the result of an inability of the ion-exchanged glass to achieve the full level of elastic strain expected from as-melted versions of the same composition. When the potassium ions invade the glass network, there is an initial elastic expansion of the local cavity. However, there is also a second aspect of the mechanical response, i.e., an irrecoverable densification as the potassium fills the “interstitial” space in the network without expansion of the network as a whole. This may be viewed as a form of plastic deformation with respect to the original as-melted glass. However, the particular mechanism of cation-induced relaxation of the network (CIRON) postulated by Ingram et al. [41], in which there is an evolution of the Qⁿ speciation of silica leading to stress reduction, is not supported by our work since we find no discernable changes in Qⁿ speciation before and after ion exchange. This constancy of Qⁿ speciation is in good agreement with the simulations of Kreski et al. [34]. Also, we note that the mechanism of shear plasticity suggested by Glebov et al. [40] is insufficient since it assumes conservation of the glass volume. In this respect, the newer concepts proposed by Varshneya [15] are clearly superior where he invokes volume plasticity and shear plasticity during ion exchange, each having a distinct yield strength.

It is yet to be resolved whether the deformations during ion exchange are more appropriately classified as *nonlinear elasticity*. One needs to examine the reversibility of the deformation with respect to changes in concentration. For example, the conversion of sodium to potassium leads to a densification that is irrecoverable on a normal time scale. However, this densification may be recoverable if the invading potassium ions are subsequently replaced again by sodium ions as a result of the fast relaxation process in the reverse direction. Complete recovery of deformations on a laboratory scale of time upon load removal (swapping the potassium ions back with sodium ions) would be better termed *nonlinear elasticity*. Incomplete recovery, on the other hand, would be termed *plasticity*. The inclusion of a slow relaxation process at increased temperatures may in fact be considered *delayed elasticity* typical of glass transformation range behavior [1]. We anticipate conducting molecular simulations using larger number of atoms in the near future to address these issues.

The network dilation anomaly may also be considered as a manifestation of the “forbidden glass” concept proposed by Mauro and Loucks [42]. In their work, Mauro and Loucks showed that there are rigid bounds in phase space to the locus of all possible glasses obtainable by cooling from the liquid state. The temperature dependence of the equilibrium liquid state is described by a well-defined one-dimensional path in phase space. For any non-instantaneous cooling path to a final glassy state, there is some departure from this one-dimensional path. The combination of all possible cooling paths from the liquid state forms a confined volume of possible glasses states in phase space. While glasses outside of this volume are theoretically possible, they are not obtainable via any temperature profile that starts from the equilibrium liquid state. These glasses are hence termed “forbidden glasses” and are only obtainable via non-fusion techniques such as vapor deposition or sol gel processing. The final glass structures obtained via ion exchange present another case of forbidden glasses, since the local environment of potassium due to the ion exchange process is not achievable through thermal processing alone.

The concept of forbidden glass is also intimately connected to the relaxation behavior of glass and the failure of the fictive temperature description of the glassy state [42,43]. Ion-exchanged glasses provide an especially clear case where the notion of fictive temperature completely fails, since it is not possible for these glasses to be described in terms of any combination of equilibrium liquid states. Nor is it possible for an ion-exchanged glass to be described in terms of any combination of as-melted glassy states. This severely complicates the modeling of structural relaxation in these glasses. An alternative concept of “configurational temperature” proposed by Araujo and Mauro [44] may provide another path forward. However, the modeling of structural relaxation in ion-exchanged glasses remains a significant challenge that has not yet been addressed in the literature.

5. Conclusions

The linear network dilation coefficient of a glass describes the amount of linear strain per unit change in alkali concentration. During the ion exchange process, the observed network dilation coefficient is significantly less than that obtained for as-melted glasses. Our molecular dynamics simulations reveal that this anomalous behavior of the LNDC is the result of different local environments for alkali ions in the as-melted vs. ion-exchanged glasses. During the ion exchange process potassium ions are stuffed into sites previously occupied by sodium ions. The relaxation of local environment around potassium is a two-

stage process. The observed LNDC is the result of the first stage of relaxation only, and therefore the full expansion desired by potassium is not allowed. The network dilation anomaly thereby reduces the practical level of compressive stress achievable in ion-exchanged glasses.

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